Stochastic Analysis

LectureNotes

Phd.Student
LaTeX by Zheng.zh
SUST University

March 25, 2026

Contents

1 Intro **2**

1.1 Set-up 3

1.2 Filtration 4

1.3 Stopping Times 5

1.4 Cts-time martingale 8

1.4.1 The Doob-Meyer Decomposition 14

1.5 Square Integrable 19

1.6 L^2Theory 22

1.7 Itó Integral 24

Chapter 1

Intro

Riemann-Stieltjes Integral: If f is continuous and g is of finite variation then $\int_a^b f(x)\ \mathrm{d}g(x)$ is well-defined and is defined to be

$$\int_a^b f(x)\ \mathrm{d}g(x) := \lim_{n\to\infty} \sum_{i=0}^{n-1} f(\xi_i)\left(g(x_{i+1}) - g(x_i)\right)$$

with $a \le x_0 < x_1 < \cdots < x_n \le b$ and $\xi_i \in [x_i, x_{i+1}]$.

Main Task: Itô integral $(H \cdot X)_t = \int_0^t H_s\ \mathrm{d}X_s$.

E.g X is Brownian motion.

Theorem 1.0.1. $\displaystyle\sum_{k=1}^{2^n} \left(B_{\frac{k}{2^n}} - B_{\frac{k-1}{2^n}}\right)^2 \xrightarrow{a.s.} 1.$

Finite Variation of a function $g : [a, b] \to \mathbb{R}$ is

$$V_g([a, b]) = \int_a^b |\mathrm{d}g(x)| = \sup_n \sum_{i=1}^{n} |g(x_{i+1}) - g(x_i)|$$

Corollary 1.0.1. $V_B([0, 1]) = \infty$ a.s.

Proof for Corollary.

■.

$$\begin{aligned}(\star) \sum_{k=1}^{2^n} \left(B_{\frac{k}{2^n}} - B_{\frac{k-1}{2^n}}\right)^2 &\le \sup_{1\le k\le 2^n} \left|B_{\frac{k}{2^n}} - B_{\frac{k-1}{2^n}}\right| \cdot \sum_{k=1}^{2^n} \left|B_{\frac{k}{2^n}} - B_{\frac{k-1}{2^n}}\right| \\ &\le V_B([0, 1]) \cdot \sup_{1\le k\le 2^n} \left|B_{\frac{k}{2^n}} - B_{\frac{k-1}{2^n}}\right|\end{aligned}$$

If $V_B([0, 1]) < \infty$, then $(\star) \to 0$.

Contradiction with $(\star) \to 1$, so w.p.1 $V_B([0, 1]) = \infty$. □

$(H \cdot X)_t = \int_0^t H_s\ \mathrm{d}X_s$, where H_s is previsible or predictable, X_s is semi martingale.

Example . Let $H_n \in \mathcal{F}_{n-1}$, (M_n) be a mart.

Then $(H \cdot M)_n = \sum_{k=1}^n H_k(M_k - M_{k-1})$ is a martingale.

1.1 Set-up

Given a probability space $(\Omega, \mathcal{F}, \mathsf{P})$. A stochastic process $X = (X_t)_{t\geq 0}$ is a collection of random variables taking values in $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$.

$$\begin{aligned} X\colon [0,\infty) \times \Omega &\to \mathbb{R}^d \\ (t,\omega) &\mapsto X_t(\omega). \end{aligned}$$

Fix ω, then the function $t \mapsto X_t(\omega)$ is called a sample path.

Definition 1.1.1. For two stochastic process X and Y, we say Y is a modification of X if

$$\forall\ t \geq 0, \quad \mathsf{P}(X_t = Y_t) = 1$$

Definition 1.1.2. We say X and Y are indistinguishable if

$$\mathsf{P}(X_t = Y_t,\ \forall\ t \geq 0) = 1.$$

And then Y is a version of X.

Example . Let Y be a modification of X and suppose X and Y both have a.s. right continuous sample paths. Then X & Y are indistinguishable.

Definition 1.1.3. A function f is right-continuous if $f(x) = f(x+) = \lim_{t\downarrow x} f(t)$.

Proof. Since $\forall\ t \geq 0$, $\mathsf{P}(X_t \neq Y_t) = 0$, we get

$$\mathsf{P}\left(\bigcup_{q\in\mathbb{Q}^+} X_q \neq Y_q\right) \leq \sum_{q\in\mathbb{Q}^+} \mathsf{P}(X_q \neq Y_q) = 0.$$

$$\Rightarrow\ \mathsf{P}\left(\bigcap_{q\in\mathbb{Q}^+} X_q = Y_q\right) = 1.$$

For two right-continuous functions f and g, if $f(q) = g(q)$, $\forall\ q \in \mathbb{Q}^+$, then $f(t) = g(t)$, $\forall\ t \geq 0$.

Hence $\mathsf{P}\left(X_t = Y_t,\ \forall\ t \geq 0\right) = 1$. □

Example . Let $T \sim \mathrm{Exp}(1)$. Define

$$\begin{aligned} X_t &\equiv 0, \quad \forall\ t \geq 0 \\ Y_t &= \begin{cases} 0, & t \neq T \\ 1, & t = T. \end{cases} \end{aligned}$$

$\forall\ t \geq 0$, $\mathsf{P}(X_t \neq Y_t) = \mathsf{P}(Y_t = 1) = \mathsf{P}(t = T) = 0$.

1.2 Filtration

Let $(\mathcal{F}_t)_{t\geq 0}$ be a filtration of $(\Omega, \mathcal{F})$ such that $\forall\ 0 \leq s < t < \infty$, $\mathcal{F}_s \subseteq \mathcal{F}_t \subseteq \mathcal{F}$.

Example . (Natural Filtration) Let $(X_t)_{t\geq 0}$ be a stochastic process, Define

$$\mathcal{F}_t^X = \sigma(X_s, 0 \leq s \leq t).$$

Then $(\mathcal{F}_t^X)$ is a filtration.

Obviously, $X_t \in \mathcal{F}_t^X$.

Exercise 1.7. Let X be a process, every sample path of which is <u>RCLL</u>. Let

$$A = \{\omega \colon X(\omega) \text{ is continuous on } [0, t_0)\}$$

for some fixed $t_0 > 0$. Show that $A \in \mathcal{F}_{t_0}^X$.

<u>RCLL</u>: $f(x) = f(x+)$ and $f(x-)$ exists.

Proof.

$$\begin{aligned} A &= \{\omega \colon \forall\ s \in (0, t_0),\ X(s-, \omega) = X(s, \omega)\} \\ &= \{\omega \colon \forall\ q \in (0, t_0) \cap \mathbb{Q},\ X(q-, \omega) = X(q, \omega)\} \end{aligned}$$

Let $(0, t_0) \cap \mathbb{Q} = (q_m)_{m\geq 1}$, then

$$\begin{aligned} A &= \left\{\omega \colon \forall\ q_m \text{ with } m \geq 1,\ \forall\ k \geq 1,\ \exists\ N \geq 1,\ \forall\ n \geq N, \left|X(q_m - \frac{1}{n}, \omega) - X(q_m, \omega)\right| < \frac{1}{k}\right\} \\ &= \bigcap_{m=1}^{\infty} \bigcap_{k=1}^{\infty} \bigcup_{N=1}^{\infty} \bigcap_{n=N}^{\infty} \left\{\left|X(q_m - \frac{1}{n}, X(q_m)) < \frac{1}{k}\right|\right\} \in \mathcal{F}_{t_0}^X. \end{aligned}$$

□

Exercise 1.8. Let X be a process whose sample paths are RCLL a.s. Then $A \notin \mathcal{F}_{t_0}^X$ is possible.

Proof. [TODO] □

Definition 1.2.1. A filtration $\{\mathcal{F}_t\}_{t\geq 0}$ is said to satisfy the <u>usual condition</u> if

(a) $\mathcal{F}_t = \mathcal{F}_{t+} = \bigcap_{s>t} \mathcal{F}_s$;

(b) $\mathcal{F}_0$ contains all the <u>$\mathsf{P}-$negligible</u> sets in $\mathcal{F}$.

Definition 1.2.2. An even A is P-negligible if $\exists\ N \in \mathcal{F}$ s.t. $A \subseteq N$, $\mathsf{P}(N) = 0$.

Definition 1.2.3. $\mathcal{F}_{t+} = \bigcap_{s>t} \mathcal{F}_s$, $\mathcal{F}_{t-} = \sigma\left(\bigcup_{s<t} \mathcal{F}_s\right)$. We say a filtration is right-continuous if $\mathcal{F}_t = \mathcal{F}_{t+}$.

Definition 1.2.4. The stochastic process X is adapted to the filtration $\{\mathcal{F}_t\}_{t\geq 0}$ if $\forall\ t \geq 0$, $X_t \in \mathcal{F}_t$.

Obviously (X_t) is adapted to the natural filtration.

Definition 1.2.5. The stochastic process X is called progressively measurable with respect to $(\mathcal{F}_t)$ if $\forall\ t \geq 0$, the map

$$(s,\omega) \mapsto X_s(\omega)$$
$$([0,t] \times \Omega, \mathcal{B}([0,t]) \otimes \mathcal{F}_t) \to (\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$$

is measurable.

adapted $\supseteq$ progressively measurable $\supseteq$ previsible

Proposition 1.2.1. If X is adapted to the filtration $(\mathcal{F}_t)$ and every sample path is right-continuous or left-continuous, then X is also progressively measurable.

Proof for Proposition.

■. Assume right-continuity $\forall\ t > 0$, $\forall\ n \geq 1$, $\forall\ 0 \leq k \leq 2^n - 1$,

$$X_s^{(n)}(\omega) = X_{\frac{k+1}{2^n}t}(\omega), \quad \forall\ \frac{kt}{2^n} < s \leq \frac{k+1}{2^n}t$$

Then $\lim\limits_{n\to\infty} X_s^n(\omega) = X_s(\omega)$ for all $(s,\omega) \in [0,t] \times \Omega$.

It suffices to prove

$$(s,\omega) \mapsto X_s^{(n)}(\omega)$$
$$([0,t] \times \Omega, \mathcal{B}([0,t]) \otimes \mathcal{F}_t) \to (\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$$

is measurable.

$\forall\ A \in \mathcal{B}(\mathbb{R}^d)$,

$$\begin{aligned}&\left\{(s,\omega)\colon X_s^{(n)}(\omega) \in A\right\}\\ =&\bigcup_{k=0}^{2^n-1}\left(\frac{k}{2^n}t, \frac{k+1}{2^n}t\right] \times \left\{X_{\frac{k+1}{2^n}t} \in A\right\} \in \mathcal{B}([0,t]) \otimes \mathcal{F}_t\end{aligned}$$

We get $X_s^{(n)}(\omega)$ is measurable. Hence $X_s = \lim_{n\to\infty} X_s^{(n)}$ is also measurable. □

1.3 Stopping Times

$(\Omega, \mathcal{F}, \mathsf{P})$ and $(\mathcal{F}_t)_{t\geq 0}$.

Definition 1.3.1. A random time T is a stopping time of $(\mathcal{F}_t)$ if $\forall\ t \geq 0$, $\{T \leq t\} \in \mathcal{F}_t$.

Definition 1.3.2. A random time T is an optional time of $(\mathcal{F}_t)$ if $\forall\ t \geq 0$, $\{T < t\} \in \mathcal{F}_t$.

Proposition 1.3.1. stopping time $\subseteq$ optional time and stopping $=$ optional if the filtration is right-continuous.

Proof for Proposition.

- "$\Rightarrow$" $\{T < t\} = \bigcup_{n=1}^{\infty}\{T \le t - \frac{1}{n}\} \in \mathcal{F}_t$.
  "$\Leftarrow$" If T is an optional time, then $\forall\ m \ge 1$,

$$\{T \le t\} = \bigcap_{n=m}^{\infty}\left\{T < t + \frac{1}{n}\right\}.$$

Because $\{T < t + \frac{1}{n}\} \in \mathcal{F}_{t+\frac{1}{n}} \subseteq \mathcal{F}_{t+\frac{1}{m}}$.

$\Rightarrow\ \{T \le t\} \in \mathcal{F}_{t+\frac{1}{m}}$

$\Rightarrow\ \{T \le t\} \in \bigcap_{m=1}^{\infty}\mathcal{F}_{t+\frac{1}{m}} = \mathcal{F}_{t^+} = \mathcal{F}_t$. □

Lemma 1.3.1. If T and S are stopping times, then $T \wedge S$, $T \vee S$, $T + S$ are stopping times.

Proof. $\forall\ t \ge 0$,

1. $\{T \wedge S \le t\} = \{T \le t\} \cup \{S \le t\} \in \mathcal{F}_t$.
2. $\{T \vee S \le t\} = \{T \le t\} \cap \{S \le t\} \in \mathcal{F}_t$.
3. $\{T + S \le t\} \in \mathcal{F}_t \Leftrightarrow \{T + S > t\} \in \mathcal{F}_t$.

$$\begin{aligned}
&\{T + S > t\}\\
=&\{T = 0,\ S > t\} \cup \{S = 0,\ T > t\} \cup \{0 < T < t,\ T + S > t\} \cup \{S > 0,\ T \ge t\}\\
\in&\mathcal{F}_t \cup \mathcal{F}_t \cup \{0 < T < t,\ T + S > t\} \cup \mathcal{F}_t.
\end{aligned}$$

It suffices to show

$$\{0 < T < t,\ T + S > t\} = \bigcup_{\substack{r \in \mathbb{Q}^+\\ 0<r<t}} \{r < T < t,\ S > t - r\}.$$

Because the right term belong $\mathcal{F}_t$.

"$\Leftarrow$ " $S > t - r > t - T$, so $S + T > t$.

"$\Rightarrow$ " $\exists\ r \in (0, T)$, $S + r > t$ since $S + T > t$.

□

Lemma 1.3.2. If $\{T_n\}_{n\ge1}$ are stopping times, then $\sup_{n\ge1} T_n$ is a stopping time.

Proof. $\forall\ t \ge 0$, $\{\sup_{n\ge1} T_n \le t\} = \bigcap_{n=1}^{\infty}\{T_n \le t\} \in \mathcal{F}_t$. □

Definition 1.3.3. The stopping σ-algebra is $\mathcal{F}_T = \{A \in \mathcal{F} \mid \forall\ t \ge 0, A \cap \{T \le t\} \in \mathcal{F}_t\}$.

Proposition 1.3.2. If $T(\omega) = t_0$ for some fixed $t_0 \ge 0$, $\forall\ \omega \in \Omega$, then $\mathcal{F}_T = \mathcal{F}_{t_0}$.

Proof. Note that $\forall\ t \geq 0$,

$$\{T \leq t\} = \begin{cases} \varnothing, & t_0 > t \\ \Omega, & t_0 \leq t \end{cases}$$

"$\mathcal{F}_{t_0} \subseteq \mathcal{F}_T$"

$\forall\ A \in \mathcal{F}_{t_0}$, $\forall\ t \geq 0$,

① $t \geq t_0$, $A \cap \{T \leq t\} = A \in \mathcal{F}_{t_0} \subseteq \mathcal{F}_t$;

② $t < t_0$, $A \cap \{T \leq t\} = \varnothing \in \mathcal{F}_t$.

Hence $A \in \mathcal{F}_T \Rightarrow\ \mathcal{F}_{t_0} \subseteq \mathcal{F}_T$.

"$\mathcal{F}_T \subseteq \mathcal{F}_{t_0}$"

$\forall\ A \in \mathcal{F}_T$, we have $A \cap \{T \leq t\} \in \mathcal{F}_t$ by definition. Let $t = t_0$, then $\{T \leq t_0\} = \Omega$, so

$$A \cap \{T \leq t_0\} \in \mathcal{F}_{t_0} \Rightarrow\ A \cap \Omega \in \mathcal{F}_{t_0} \Rightarrow\ A \in \mathcal{F}_{t_0}.$$

Hence $A \in \mathcal{F}_{t_0} \Rightarrow\ \mathcal{F}_T \subseteq \mathcal{F}_{t_0}$ □

Lemma 1.3.3. $\forall$ stopping time T and S, $\forall\ A \in \mathcal{F}_S$, we have $A \cap \{S \leq T\} \in \mathcal{F}_T$. In particular, if $S \leq T$ for all $\omega \in \Omega$, then $\mathcal{F}_S \subseteq \mathcal{F}_T$.

Proof. $T \wedge t$ is also a stopping time. $\{T \wedge t \leq s\} \in \mathcal{F}_s \subseteq \mathcal{F}_t$. Hence $T \wedge t \in \mathcal{F}_t$.

To prove $A \cap \{S \leq T\} \in \mathcal{F}_T$, we need to prove $\forall\ t \geq 0$, $A \cap \{S \leq T\} \cap \{T \leq t\} \in \mathcal{F}_t$.

We have

$$A \cap \{S \leq T\} \cap \{T \leq t\} = \underbrace{A \cap \{S \leq t\}}_{\in \mathcal{F}_t} \cap \underbrace{\{T \leq t\}}_{\in \mathcal{F}_t} \cap \underbrace{\{S \wedge t \leq T \wedge t\}}_{\mathcal{F}_t}.$$

($S \leq T$ on $S \leq t$, $T \leq t \Leftrightarrow \{S \wedge t \leq T \wedge t\}$) □

Lemma 1.3.4. Let S, T be two stopping times. Then $\mathcal{F}_{T \wedge S} = \mathcal{F}_T \cap \mathcal{F}_S$ and $\{T < S\}$, $\{S < T\}$, $\{T \leq S\}$, $\{S \leq T\}$, $\{T = S\}$ belong to $\mathcal{F}_S \cap \mathcal{F}_T = \mathcal{F}_{S \wedge T}$.

Proof. $\mathcal{F}_{T \wedge S} \subseteq \mathcal{F}_S$ and $\mathcal{F}_{T \wedge S} \subseteq \mathcal{F}_T \Rightarrow\ \mathcal{F}_{T \wedge S} \subseteq \mathcal{F}_T \cap \mathcal{F}_S$.

To prove $\mathcal{F}_T \cap \mathcal{F}_S \subseteq \mathcal{F}_{T \wedge S}$, $\forall\ A \in \mathcal{F}_T \cap \mathcal{F}_S$, we will prove $\forall\ t \geq 0$, $A \cap \{T \wedge S \leq t\} \in \mathcal{F}_t$.

$$\begin{aligned} A \cap \{T \wedge S \leq t\} &= A \cap (\{T \leq t\} \cup \{S \leq t\}) \\ &= (A \cap \{T \leq t\}) \cup (A \cap \{S \leq t\}) \in \mathcal{F}_t. \end{aligned}$$

Next $\{S \leq T\} \in \mathcal{F}_T$, $\{S > t\} \in \mathcal{F}_T$. It remains to prove $\{S < T\} \in \mathcal{F}_T$.

Assuming this, by interchanging T and S, we get $\{T \leq S\}$, $\{T > S\}$, $\{T < S\} \in \mathcal{F}_S$.

Then $\{S \leq T\} \in \mathcal{F}_T$, $\{S > T\}^{\complement} \in \mathcal{F}_S$.

$\Rightarrow\ \{S \leq T\} \in \mathcal{F}_S \cap \mathcal{F}_T$.

$\{T = S\} = \{S \leq T\} \cap \{T \leq S\} \in \mathcal{F}_S \cap \mathcal{F}_T$.

To prove $\{S < T\} \in \mathcal{F}_T$, we define $R = S \wedge T$ such that $\{S < T\} = \{R < T\}$. Notice that $R \in \mathcal{F}_R \subseteq \mathcal{F}_T$. Hence $\{R < T\} \in \mathcal{F}_T$. □

Summary:

(i) $T \in \mathcal{F}_T$;

(ii) If $S \leq T$, $\mathcal{F}_S \subseteq \mathcal{F}_T$;

(iii) $T \wedge t \in \mathcal{F}_{T\wedge t} \subseteq \mathcal{F}_t$;

(iv) $\mathcal{F}_{T\wedge S} = \mathcal{F}_T \cap \mathcal{F}_S$.

For each $\omega \in \Omega$, $T(\omega) \in [0, \infty)$ and $X_t(\omega) \in \mathbb{R}^d$, $\forall\ t \geq 0$, we define $X_T(\omega) = X_{T(\omega)}(\omega)$.

Proposition 1.3.3. Let $\{X_t\}_{t\geq 0}$ be a progressively measurable process, Let T be a stopping time. Then X_T defined on $\{T < \infty\}$ is $\mathcal{F}_T$-measurable and the stopped process $\{X_{T\wedge t}\}_{t\geq 0}$ is progressively measurable.

Proof. $\forall\ B \in \mathcal{B}(\mathbb{R}^d)$, $\{X_T \in B\} \in \mathcal{F}_T \Leftrightarrow \forall\ t \geq 0$, $\underbrace{\{X_T \in B\} \cap \{T \leq t\}}_{=\{X_{T\wedge t} \in B\} \cap \{T \leq t\}} \in \mathcal{F}_t$.

We only need to prove $X_{T\wedge t} \in \mathcal{F}_t$.
Observe $(s, \omega) \to (T(\omega) \wedge s, \omega)$ is $\mathcal{B}([0, t]) \otimes \mathcal{F}_t$-measurable.
By assumption, $(s, \omega) \mapsto X_s(\omega)$ is $\mathcal{B}([0, t]) \otimes \mathcal{F}_t$-measurable.
So $(s, \omega) \to (T(\omega) \wedge s, \omega) \to (X_{T(\omega)\wedge s}, \omega)$ is $\mathcal{B}([0, t]) \otimes \mathcal{F}_t$-measurable. □

1.4 Cts-time martingale

Definition 1.4.1. For $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$, a martingale is an adapted stochastic process $(X_t)_{t\geq 0}$ such that

$$\begin{cases} (i)\ \forall\ t \geq 0,\ \mathbb{E}|X_t| < \infty \\ (ii)\ \forall\ t \geq s \geq 0,\ \mathbb{E}(X_t \mid \mathcal{F}_s) = X_s. \end{cases}$$

If $\mathbb{E}(X_t \mid \mathcal{F}_s) \geq X_s$, we call it a submartingale.
If $\mathbb{E}(X_t \mid \mathcal{F}_s) \leq X_s$, we call it a supmartingale.

Proposition 1.4.1. Let $\varphi : \mathbb{R} \to \mathbb{R}$ be a convex (concave up) function. If $(X_t)_{t\geq 0}$ is a martingale, then $(\varphi(X_t))_{t\geq 0}$ is a submartingale.

E.g. $|X_t|$, X_t^2, $|X_t|^p$ for all $p \geq 1$, $(X_t)^+$ are all submartingale.
We assume $\mathbb{E}\,|\varphi(X_t)| < \infty$, $\forall\ t \geq 0$.

Proof. Jensen's inequality. □

Theorem 1.4.1. Continuous-time Doob's upcrossing inequality

$$
\begin{aligned}
T_b^{(0)} &= -\infty, \\
T_a^{(1)} &= \inf\{t \geq (T_b^{(0)})^+ \colon X_t \leq a\}, \\
T_b^{(1)} &= \inf\{t \geq T_a^{(1)} \colon X_t \geq b\}.
\end{aligned}
$$

$\forall\ l \geq 2$,

$$
\begin{aligned}
T_a^{(l)} &= \inf\{t \geq T_b^{(l-1)} \colon X_t \leq a\}, \\
T_b^{l} &= \inf\{t \geq T_a^{l} \colon X_t \geq b\}
\end{aligned}
$$

Define $U_{ab}^X[0,n] = \max\{k \geq 0 \colon T_b^{(k)} \leq n\}$ to be the # of upcrossing in an interval [0,n].

Theorem 1.4.2. Let $(X_t)_{t\geq 0}$ be a right-continuous submartingale. For all $T > 0$,

$$
\mathbb{E}\left(U_{ab}^X[0,T]\right) \leq \frac{\mathbb{E}(X_T - a)^+}{b-a}
$$

Proof. For each $n \geq 1$, we let $D_n = \mathbb{Z}/2^n = \{0, \frac{\pm 1}{2^n}, \frac{\pm 2}{2^n}, \cdots\}$.

Consider the upcrossing of X on $[0,T] \cap D_n$, $\left\{\frac{X_j}{2^n}\right\}_{j=0}^{\infty}$ is a discrete-time martingale. Then $U_{ab}^X\left([0,T] \cap D_n\right) \leq U_{ab}^X[0,T]$, $U_{ab}^X\left([0,T] \cap D_n\right) \uparrow U_{ab}^X([0,T])$.

By using discrete-time upcrossing

$$
\mathbb{E}\left(U_{ab}^X\left([0,T] \cap D_n\right)\right) \leq \frac{1}{b-a}\mathbb{E}(X_{T_n} - a)^+,
$$

where $T_n = \frac{[T \cdot 2^n]}{2^n}$. Let $n \to \infty$ to get

$$
\mathbb{E}\left(U_{ab}^X([0,T])\right) \leq \frac{1}{b-a}\mathbb{E}(X_T - a)^+.
$$

□

Proposition 1.4.2. If $(X_t)_{t\geq 0}$ is a right continuous sub-martingale, and $\sup_{t\geq 0} \mathbb{E}X_t^+ < \infty$, then $\exists\ X \in L^1$ s.t. $X_t \xrightarrow{a.s.} X$ as $t \to \infty$.

Proof.

$$
\begin{aligned}
\mathbb{E}\left(U_{ab}^X([0,\infty))\right) &= \lim_{n\to\infty} \mathbb{E}\left(U_{ab}^X([0,n])\right) \\
&\leq \limsup_{n\to\infty} \frac{1}{b-a}\mathbb{E}(X_n - a)^+ \\
&\leq \limsup_{n\to\infty} \frac{1}{b-a}\left(\mathbb{E}(X_n^+) + |a|\right) < \infty.
\end{aligned}
$$

Hence $\forall\ a < b$, $U_{ab}^X[0,\infty) < \infty$ a.s.

$\Rightarrow$ w.p.1, $\forall\ a < b \in \mathbb{Q}$, $U_{ab}^X[0,\infty) < \infty$.

Hence $\liminf_{t\to\infty} X_t \leq a \leq b < \limsup_{t\to\infty} X_t$ does not occur.

$\Leftrightarrow \liminf_{t\to\infty} X_t = \limsup_{t\to\infty} X_t$. □

Definition 1.4.2. A family of random variables $(X_t)_{t\geq 0}$ is uniformly integrable (U.I.) if

$$\lim_{M\to\infty} \sup_{t\geq 0} \mathbb{E}\left(|X_t| \, 1_{|X_t|>M}\right) = 0.$$

Remark. If $\sup_{t\geq 0} \mathbb{E}\,|X_t|^p < \infty$ for some $p > 1$, then U.I.

Theorem 1.4.3. If $X_n \xrightarrow{P} X$, then equivalent

$$\begin{cases} X_n \xrightarrow{L^1} X, \\ \mathbb{E}\,|X_n| \to \mathbb{E}\,|X|\,, \\ \{X_n\} \text{ is U.I.}. \end{cases}$$

Proposition 1.4.3. Let $Z \in L^1$. Then $\{\mathbb{E}(Z \mid \mathcal{G})\colon \mathcal{G} \text{ is a sub } \sigma\text{-field of } \mathcal{F}\}$ is U.I.

In particular $\{\mathbb{E}(Z \mid \mathcal{F}_t)\}_{t\geq 0}$ is U.I. and it is a martingale. (By Tower's property)

Proposition 1.4.4. A right continuous martingale $(X_t)_{t\geq 0}$ is U.I. if and only if $\exists\, X_\infty \in L^1$ such that $X_t = \mathbb{E}(X_\infty \mid \mathcal{F}_t)$.

Proof. "$\Rightarrow$" U.I. implies $\sup_{t\geq 0} \mathbb{E}\,|X_t| < \infty$.

Then the martingale convergence theorem implies that $\exists\, X_\infty$ s.t. $X_n \xrightarrow{a.s.} X$ and $\mathbb{E}\,|X_\infty| = \mathbb{E}\left(\lim_{t\to\infty} |X_t|\right) \leq \liminf_{t\to\infty} \mathbb{E}\,|X_t| < \infty$.

To prove $\forall\, t \geq 0$, $X_t = \mathbb{E}(X_\infty \mid \mathcal{F}_t) \Leftrightarrow \forall\, A \in \mathcal{F}_t$, $\mathbb{E}(X_t 1_A) = \mathbb{E}(X_\infty 1_A)$.

For any $s \geq 0$, we have $\mathbb{E}(X_{t+s} \mid \mathcal{F}_t) = X_t \implies \mathbb{E}(X_{t+s} 1_A) = \mathbb{E}(X_t 1_A)$

$$|\mathbb{E}(X_{t+s} 1_A) - \mathbb{E}(X_\infty 1_A)| \leq \mathbb{E}\,|X_{t+s} - X_\infty|\,|1_A| \leq \mathbb{E}\,|X_{t+s} - X_\infty| \to 0.$$

Let $s \to \infty$, $\mathbb{E}(X_{t+s} 1_A) \xrightarrow{a.s.} \mathbb{E}(X_\infty 1_A)$.

Recall U.I. + a.s. $\implies L^1$ convergence.

"$\Leftarrow$" If $\exists\, X_\infty \in L^1$ s.t. $X_t = \mathbb{E}(X_\infty \mid \mathcal{F}_t)$, then $\{X_t\}_{t\geq 0}$ is U.I. $\square$

Theorem 1.4.4 (Optional sampling Theorem). Let $(X_t)_{t\geq 0}$ be a right continuous martingale. Let $S \leq T$ be two stopping time. Suppose either

① $\{X_t\}$ U.I.,

② $\exists$ constant $K > 0$, $S, T \leq K$.

Then $\mathbb{E}(X_T \mid \mathcal{F}_S) = X_S \implies \mathbb{E}(X_T) = \mathbb{E}(X_S) = \mathbb{E}(X_0)$.

Proof. Define ① $Z = X_\infty$, ② $Z = X_K$.

In either case, $X_t = \mathbb{E}(Z \mid \mathcal{F}_t)$, ① $\forall\, t \geq 0$, ② $\forall\, t \leq K$.

It suffices to prove $X_T = \mathbb{E}(Z \mid \mathcal{F}_T)$. Then $X_S = \mathbb{E}(Z \mid \mathcal{F}_S)$.

Hence

$$\begin{aligned}\mathbb{E}(X_T \mid \mathcal{F}_S) &= \mathbb{E}\left(\mathbb{E}(Z \mid \mathcal{F}_T) \mid \mathcal{F}_S\right) \\ &= \mathbb{E}(Z \mid \mathcal{F}_S) \\ &= X_S.\end{aligned}$$

① Suppose $T \in \{t_1, t_2, \dots\}$. Then $\forall\ A \in \mathcal{F}_T$,

$$\begin{aligned}\mathbb{E}(X_T 1_A) &= \mathbb{E}(Z 1_A) \\ &= \sum_{n=1}^{\infty} \mathbb{E}(X_{t_n} 1_{\{T=t_n\}} 1_A) \\ &= \sum_{n=1}^{\infty} \mathbb{E}\left(1_{A \cap \{T=t_n\}} Z\right) \\ &= \mathbb{E}(1_A Z)\end{aligned}$$

because ① $A \cap \{T = t_n\} \in \mathcal{F}_{t_n}$, ② $\mathbb{E}(Z \mid \mathcal{F}_{t_n}) = X_{t_n}$

② General T. Define $T_k = \frac{[2^k T]+1}{2^k}$.

To prove $X_T = \mathbb{E}(Z \mid \mathcal{F}_T)$, $\forall\ A \in \mathcal{F}_T \subseteq \mathcal{F}_{T_k}$, we get $\mathbb{E}(Z \mid \mathcal{F}_{T_k}) = X_{T_k} \implies \mathbb{E}(Z 1_A) = \mathbb{E}(X_{T_k} 1_A)$.

Send $k \to \infty$ to see $\mathbb{E}(Z 1_A) = \mathbb{E}(X_T 1_A)$.

Because ① $X_{T_k} \xrightarrow{a.s.} X_T$ by right-continuity, ② $\{X_{T_k}\}$ is U.I. by $X_{T_k} = \mathbb{E}(Z \mid \mathcal{F}_{T_k})$, ③ $\implies X_{T_k} \xrightarrow{L^1} X_T$. □

Theorem 1.4.5 (Doob's Maximal Inequality)**.** Let $(X_t)_{t \geq 0}$ be a right-continuous submartingale. Then $\forall\ \lambda > 0$,

$$\begin{aligned}\mathsf{P}(\sup_{0 \leq s \leq t} X_s > \lambda) &\leq \frac{1}{\lambda} \mathbb{E} X_t^+, \\ \mathsf{P}(\inf_{0 \leq s \leq t} X_s < -\lambda) &\leq \frac{1}{\lambda}(\mathbb{E} X_t^+ - \mathbb{E} X_0).\end{aligned}$$

Proof. $\forall\ \lambda > 0$, $T = \inf\{t \geq 0 : X_t > \lambda\}$, then

$$\mathsf{P}(\sup_{0 \leq s \leq t} X_s > \lambda) = \mathsf{P}(T \leq t).$$

Consider $\{X_{t \wedge T}, t \geq 0\}$,

$$\begin{aligned}\lambda \mathsf{P}(T \leq t) = \lambda \mathbb{E}(1_{T \leq t}) &\leq \mathbb{E}(X_T 1_{T \leq t}) \\ &= \mathbb{E}(X_{T \wedge t} 1_{T \leq t}) \\ &= \mathbb{E}(X_{T \wedge t}^+ 1_{T \leq t}) \\ &= \mathbb{E}(X_{T \wedge t}^+) \leq \mathbb{E}(X_t^+).\end{aligned}$$

Define $S = \inf\{t \geq 0 : X_t < -\lambda\}$, then

$$\mathsf{P}(\inf_{0 \leq s \leq t} X_s < -\lambda) = \mathsf{P}(S \leq t).$$

and

$$\lambda \mathsf{P}(S \leq t) = \lambda \mathbb{E}(1_{S \leq t}).$$

Because $X_S \leq -\lambda$, $\lambda \leq -X_S$, so

$$\lambda \mathsf{P}(S \leq t) \leq \mathbb{E}(-X_S 1_{S \leq t}) = -\mathbb{E}(X_{S \wedge t} 1_{S \leq t}).$$

By submartingale optional stopping

$$\begin{aligned}
\mathbb{E}X_0 \leq \mathbb{E}X_{t \wedge S} &= \mathbb{E}(X_{t \wedge S} 1_{S \leq t}) + \mathbb{E}(X_{t \wedge S} 1_{S > t}) \\
&\leq -\lambda \mathsf{P}(S \leq t) + \mathbb{E}(X_{t \wedge S}^+) \\
&\leq -\lambda \mathsf{P}(S \leq t) + \mathbb{E}(X_t^+).
\end{aligned}$$

□

Theorem 1.4.6. $\forall\, p > 1$, $\mathbb{E}\left(\sup_{s \leq t} |X_s|\right)^p \leq \left(\frac{p}{p-1}\right)^p \mathbb{E}\,|X_t|^p$.

Proof. Define $Y = \sup_{0 \leq s \leq t} |X_s|$.

$$\begin{aligned}
\lambda \mathsf{P}(Y \geq \lambda) &\leq \mathbb{E}(|X_{T \wedge t}|\, 1_{T \leq t}) \\
&= \mathbb{E}\left(|X_{T \wedge t}|\right) - \mathbb{E}\left(|X_{T \wedge t}|\, 1_{T > t}\right) \\
&\leq \mathbb{E}\left(|X_t|\right) - \mathbb{E}\left(|X_t|\, 1_{T > t}\right) \\
&= \mathbb{E}\,|X_t| - \mathbb{E}\left(|X_t|\, 1_{Y < \lambda}\right) \\
&= \mathbb{E}\left(|X_t|\, 1_{Y \geq \lambda}\right).
\end{aligned}$$

Hence $\mathsf{P}(Y \geq \lambda) \leq \frac{1}{\lambda}\mathbb{E}\left(|X_t| \, 1_{Y \geq \lambda}\right)$.

$$\begin{aligned}
\mathbb{E}Y^p &= \int_0^\infty p\lambda^{p-1}\mathsf{P}(Y \geq \lambda) \; d\lambda \\
&\leq \int_0^\infty p\lambda^{p-2}\mathbb{E}\left(|X_t| \, 1_{Y \geq \lambda}\right) \; d\lambda \\
&= \mathbb{E}\left(|X_t| \int_0^\infty p\lambda^{p-2} 1_{Y \geq \lambda} \; d\lambda\right) \\
&= \mathbb{E}\left(|X_t| \int_0^Y p\lambda^{p-2} \; d\lambda\right) \\
&= \mathbb{E}\left(|X_t| \, \frac{p}{p-1} Y^{p-1}\right) \\
&\leq \frac{p}{p-1}\left(\mathbb{E}\,|X_t|^p\right)^{\frac{1}{p}} \left(\mathbb{E}(Y^{p-1})^q\right)^{\frac{1}{q}} \\
&= \frac{p}{p-1}\left(\mathbb{E}\,|X_t|^p\right)^{\frac{1}{p}} \left(\mathbb{E}Y^p\right)^{\frac{1}{q}}.
\end{aligned}$$

$\implies (\mathbb{E}Y^p)^{\frac{1}{p}} \leq \frac{p}{p-1}\left(\mathbb{E}\,|X_t|^p\right)^{\frac{1}{p}}$.

Consider $Y \wedge k$, $\forall\, k > 0$,

$$\begin{aligned}
\mathbb{E}\,(Y \wedge k)^p &= \int_0^\infty p\lambda^{p-1}\mathsf{P}(Y \wedge k \geq \lambda) \; d\lambda \\
&= \int_0^k p\lambda^{p-1}\mathsf{P}(Y \wedge k \geq \lambda) \; d\lambda + \underbrace{\int_k^\infty p\lambda^{p-1}\mathsf{P}(Y \wedge k \geq \lambda) \; d\lambda}_{=0} \\
&= \int_0^k p\lambda^{p-1}\mathsf{P}(Y \wedge k \geq \lambda) \; d\lambda \\
&= \int_0^k p\lambda^{p-1}\mathsf{P}(Y \geq \lambda) \; d\lambda \\
&\leq \mathbb{E}\left(|X_t| \int_0^k p\lambda^{p-1} 1_{Y \geq \lambda} \; d\lambda\right) \\
&= \mathbb{E}\left(|X_t| \int_0^{k \wedge Y} p\lambda^{p-2} \; d\lambda\right) \\
&= \mathbb{E}\left[|X_t| \, \frac{p}{p-1}(Y \wedge k)^{p-1}\right].
\end{aligned}$$

$\implies \mathbb{E}(Y \wedge k)^p \leq \frac{p}{p-1}\left(\mathbb{E}\,|X_t|^p\right)^{\frac{1}{p}} \left(\mathbb{E}(Y \wedge k)^p\right)^{\frac{1}{q}}$.

$\implies (\mathbb{E}(Y \wedge k)^p)^{\frac{1}{p}} \leq \frac{p}{p-1}\left(\mathbb{E}\,|X_t|^p\right)^{\frac{1}{p}}$.

Let $k \to \infty$, we get

$$(\mathbb{E}Y^p)^{\frac{1}{p}} \leq \frac{p}{p-1}\left(\mathbb{E}\,|X_t|^p\right)^{\frac{1}{p}}.$$

$\square$

1.4.1 The Doob-Meyer Decomposition

Recall the Doob Decomposition

$$X_n = M_n + A_n,$$

where X_n is a submartingale, M_n is a martingale, and A_n is a predictable increasing process.

$$\begin{aligned}
&\mathbb{E}\left(M_{n+1} \mid \mathcal{F}_n\right) = M_n \\
\Longrightarrow\ &\mathbb{E}(X_{n+1} - A_{n+1} \mid \mathcal{F}_n) = X_n - A_n \\
\Longrightarrow\ &\mathbb{E}(X_{n+1} \mid \mathcal{F}_n) - A_{n+1} = X_n - A_n \\
\Longrightarrow\ &A_{n+1} - A_n = \mathbb{E}(X_{n+1} \mid \mathcal{F}_n) - X_n \geq 0.
\end{aligned}$$

$$\begin{cases} A_n = \sum_{k=0}^{n-1} \left(\mathbb{E}(X_{k+1} \mid \mathcal{F}_k) - X_k\right) \\ A_0 = 0. \end{cases}$$

Next, for a right-continuous submartingale $(X_t)_{t\geq 0}$, $X_t = M_t + A_t$, where M_t is a martingale, A_t is a increasing & "predictable".

Consider $(A_t)_{t\geq 0}$ adapted.

Definition 1.4.3. An adapted process A is called increasing if a.s.

(a) $A_0 = 0$;

(b) $t \mapsto A_t$ is nondecreasing and right-continuous;

(c) $\mathbb{E}(A_t) < \infty$, $\forall\ t \geq 0$.

Integrable: $\mathbb{E}(A_\infty) = \mathbb{E}(\lim_{t\to\infty} A_t) < \infty$.

Definition 1.4.4. An increasing process A is called natural if $\forall$ bounded, right-continuous martingale $(M_t)_{t\geq 0}$, we have $\forall\ t > 0$,

$$\mathbb{E}\int_0^t M_s\ dA_s = \mathbb{E}\int_0^t M_{s-}\ dA_s.$$

Remark. If $s \mapsto A_s$ is a.s. continuous, then A is natural.

Definition 1.4.5. An increasing sequence $(A_n, \mathcal{F}_n)$ is called natural if $\forall$ bounded martingale $\{M_n, \mathcal{F}_n\}$,

$$\mathbb{E}(M_n A_n) = \mathbb{E}\left(\sum_{k=1}^{n} M_{k-1}(A_k - A_{k-1})\right).$$

Proposition 1.4.5. A predictable $\Longleftrightarrow$ A natural.

Proof. "$\Rightarrow$ " To prove

$$\begin{aligned}
&\mathbb{E}(M_nA_n) = \sum_{k=1}^{n}\mathbb{E}M_{k-1}(A_k - A_{k-1}) - \sum_{k=1}^{n}(M_{k-1}A_k) - \mathbb{E}(M_{k-1}A_{k-1}) \\
\iff &\sum_{k=2}^{n}\mathbb{E}(M_{k-1}A_{k-1}) + \mathbb{E}(M_nA_n) = \sum_{k=1}^{n}\mathbb{E}(M_{k-1}A_k) \\
\iff &\sum_{k=1}^{n}\mathbb{E}(M_kA_k) = \sum_{k=1}^{n}\mathbb{E}(M_{k-1}A_k) \\
\iff &\sum_{k=1}^{n}\mathbb{E}\left[(M_k - M_{k-1})A_k = 0.\right]
\end{aligned}$$

So if $A_n \in \mathcal{F}_{n-1}$,

$$\mathbb{E}\left(A_n(M_n - M_{n-1})\right) = \mathbb{E}\left[\underbrace{\mathbb{E}(M_n - M_{n-1} \mid \mathcal{F}_n)}_{=0} A_n\right] = 0.$$

"$\Leftarrow$ " Assume A is natural. Then $\forall$ bounded (M_n),

$$\mathbb{E}\left[A_n(M_n - M_{n-1})\right] = 0, \quad \forall\ n \geq 1.$$

Goal: $A_n \in \mathcal{F}_{n-1} \iff A_n = \mathbb{E}(A_n \mid \mathcal{F}_{n-1})$ a.s..

$$\begin{aligned}
&\mathbb{E}\left[M_n\left(A_n - \mathbb{E}(A_n \mid \mathcal{F}_n)\right)\right] \\
=&\mathbb{E}\left(A_nM_{n-1}\right) - \mathbb{E}\left[M_n\mathbb{E}(A_n \mid \mathcal{F}_{n-1})\right] \\
=&\mathbb{E}\left(M_{n-1}\left(A_n - \mathbb{E}(A_n \mid \mathcal{F}_{n-1})\right)\right) + \mathbb{E}\left[\mathbb{E}(A_n \mid \mathcal{F}_{n-1})(M_n - M_{n-1})\right] \\
=&\mathbb{E}(M_{n-1}A_n) - \mathbb{E}\left[\mathbb{E}(A_nM_{n-1} \mid \mathcal{F}_{n-1})\right] + \mathbb{E}\left[\mathbb{E}(A_n \mid \mathcal{F}_{n-1})\mathbb{E}(M_n - M_{n-1} \mid \mathcal{F}_{n-1})\right] = 0.
\end{aligned}$$

Fix $n \geq 1$. Let

$$X = \operatorname{sgn}\left(A_n - \mathbb{E}(A_n \mid \mathcal{F}_{n-1})\right)$$

Then $X\left(A_n - \mathbb{E}(A_n \mid \mathcal{F}_{n-1})\right) = |A_n - \mathbb{E}(A_n \mid \mathcal{F}_{n-1})|$.

Define

$$M_k = \begin{cases} \mathbb{E}(X \mid \mathcal{F}_k), & 0 \leq k \leq n-1 \\ X, & k = n \end{cases}$$

Check: $\mathbb{E}(M_{k+1} \mid \mathcal{F}_k) = M_k$, $k \geq n$, $\mathbb{E}(X \mid \mathcal{F}_k) = X$.

$$\begin{aligned}
0 &= \mathbb{E}\left[M_n(A_n - \mathbb{E}(A_n \mid \mathcal{F}_{n-1}))\right] \\
&= \mathbb{E}\left|A_n - \mathbb{E}(A_n \mid \mathcal{F}_{n-1})\right|
\end{aligned}$$

$\implies A_n = \mathbb{E}(A_n \mid \mathcal{F}_{n-1})$ a.s.

$\implies A_n \in \mathcal{F}_{n-1}$. $\square$

$$\mathbb{E}(M_nA_n) = \sum_{k=1}^{n} \mathbb{E}\left(M_{k-1}(A_k - A_{k-1})\right)$$
$$\Longleftrightarrow \mathbb{E}(M_tA_t) = \mathbb{E}\left(\int_0^t M_{s-}\ dA_s\right) = \mathbb{E}\left(\int_0^t M_s\ dA_s\right).$$
$$\int_0^t M_s\ dA_s = \lim_{\|\Pi\|\to 0} \int_0^t M_s^{\Pi}\ dA_s.$$

where $\Pi = \{t_0, t_1, \cdots, t_n\} \subseteq [0,t]$ with $0 = t_0 < t_1 < \cdots < t_n = t$ and $\|\Pi\| = \max_{1\le k\le n}(t_k - t_{k-1})$. $M_s \xleftarrow{a.s.} M_s^{\Pi} = \sum_{k=1}^{n} M_{t_k} \mathbf{1}_{(t_{k-1},t_k]}(s)$ by right-continuity. For each Π,

$$\begin{aligned}
&\mathbb{E}\int_0^t M_s^{\Pi}\ dA_s \\
=&\mathbb{E}\sum_{k=1}^{n} M_{t_k}(A_{t_k} - A_{t_{k-1}}) \\
=&\sum_{k=1}^{n} \mathbb{E}(M_{t_kA_{t_k}}) - \mathbb{E}(M_{t_k}A_{t_{k-1}}) \\
=&\mathbb{E}(M_{t_n}A_{t_n}) + \sum_{k=1}^{n-1} \mathbb{E}(M_{t_k}A_{t_k}) - \sum_{k=2}^{n} \mathbb{E}(M_{t_k}A_{t_{k-1}}) \\
=&\mathbb{E}(M_tA_t) + \sum_{k=1}^{n-1} \mathbb{E}(M_{t_k}A_{t_k}) - \sum_{k=1}^{n-1} \mathbb{E}(M_{t_{k+1}}A_{t_k}) \\
=&\mathbb{E}(M_tA_t) - \underbrace{\sum_{k=1}^{n} \mathbb{E}\left(A_{t_k}(M_{t_{k+1}} - M_{t_k})\right)}_{=0}
\end{aligned}$$

Definition 1.4.6. Let $S = \{\text{stopping time } T \text{ with } T < \infty \text{ a.s.}\}$. We say a right-continuous $\{X_t\}_{t\ge 0}$ is of class D if $\{X_T\}_{T\in S}$ is U.I. $\Longrightarrow$ implies $\{X_t\}_{t\ge 0}$ is U.I..

Fix $a > 0$. Let $S_a = \{\text{stopping time } T \text{ with } T \le a \text{ a.s.}\}$. $\{X_t\}_{t\ge 0}$ is of class DL if $\{X_T\}_{T\in S_a}$ is U.I. for every $a > 0$.

Theorem 1.4.7 (Doob-Meyer Decomposition)**.** If $\{X_t\}$ is of class DL, then $X_t = M_t + A_t$ with $\{M_t\}$ is martingale, $\{A_t\}$ is increasing.

If (A_t) is natural, then (M_t, A_t) is unique.

Proof. Assume $X_t = \begin{cases} M_t + A_t \\ M_t' + A_t' \end{cases}$ with M_t, M_t' martingale, A_t, A_t' natural.

Let $B_t = M_t - M_t' = A_t' - A_t$ is a martingale and of bounded variation <u>natural</u>.

$\forall$ bounded continuous martingale $\{\xi_t\}_{t\geq 0}$,

$$\begin{aligned}\mathbb{E}\xi_t(A_t - A'_t) &= \mathbb{E}\int_0^t \xi_{s-}(dA_s - dA'_s)\\ &= \mathbb{E}\left[\int_0^t \xi_{s-}dB_s\right]\\ &= \lim_{n\to\infty}\mathbb{E}\sum_{j=1}^{m_n}\xi_{t_{j-1}^{(n)}}\left[B_{t_j^{(n)}} - B_{t_{j-1}^{(n)}}\right]\end{aligned}$$

where $\Pi_n = \{t_0^{(n)}, \cdots, t_{m_n}^{(n)}\}$ is a partition of $[0,t]$.

$$\mathbb{E}\left[\xi_{t_{j-1}^{(n)}}(B_{t_j^{(n)}} - B_{t_{j-1}^{(n)}})\right] = \mathbb{E}\left[\xi_{t_{j-1}^{(n)}}\underbrace{\mathbb{E}(B_{t_j^{(n)}} - B_{t_{j-1}^{(n)}} \mid \mathcal{F}_{t_{j-1}^{(n)}})}_{=0}\right]$$

Hence $\mathbb{E}\left[\xi_t(A_t - A'_t)\right] = 0$.

$\forall$ bounded variable ξ, we let $\xi_t = \mathbb{E}(\xi \mid \mathcal{F}_t)$. Then $\mathbb{E}\left[\mathbb{E}(\xi \mid \mathcal{F}_t)(A_t - A'_t)\right] = 0 \Rightarrow \mathbb{E}\left[\xi(A_t - A'_t)\right] = 0$.

$\xi = \operatorname{sgn}(A_t - A'_t) \Rightarrow \mathbb{E}\left|A_t - A'_t\right| = 0 \Rightarrow A_t = A'_t$, a.s. $\forall\ t > 0 \Rightarrow A_t = A'_t$, $\forall\ q \in \mathbb{Q}_+$ a.s. $\Rightarrow$ Right-continuity, $A_t = A'_t$, $\forall\ t \geq 0$ a.s..

Existence: Fix $a > 0$.

We only need to prove $X_t = M_t + A_t$, $\forall\ 0 \leq t \leq a$.

Consider $Y_t \triangleq X_t - \mathbb{E}(X_a \mid \mathcal{F}_t)$. $(Y_t)_{t\geq 0}$ is a submartingale because

$$\begin{aligned}\mathbb{E}(Y_t \mid \mathcal{F}_s) &= \mathbb{E}(X_t \mid \mathcal{F}_s) - \mathbb{E}(X_a \mid \mathcal{F}_s)\\ &\geq X_s - \mathbb{E}(X_a \mid \mathcal{F}_s) = Y_s, \quad \forall\ t \geq s.\end{aligned}$$

$\Pi_n = \{t_0^{(n)}, \cdots, t_{2^n}^{(n)}\}$ with $t_j^{(n)} = \frac{j}{2^n}a$, $0 \leq j \leq 2^n$. Then $\forall\ 0 \leq j \leq 2^n$,

$$Y_{t_j^{(n)}} = M_{t_j^{(n)}}^{(n)} + A_{t_j^{(n)}}^{(n)}$$

with $A_{t_j^{(n)}}^{(n)} = \sum_{k=0}^{j-1}\mathbb{E}\left(Y_{t_{k+1}^{(n)}} - Y_{t_k^{(n)}}\middle|\mathcal{F}_{t_k^{(n)}}\right)$

$$\begin{cases}Y_0 = X_0 - \mathbb{E}(X_a \mid \mathcal{F}_\prime) \leq 0\\ Y_a = X_a - \mathbb{E}(X_a \mid \mathcal{F}_a) = 0.\end{cases}$$

$0 = M_a^{(n)} - A_a^{(n)}$. Hence $M_{t_j^{(n)}}^{(n)} = \mathbb{E}(M_a^{(n)} \mid \mathcal{F}_{t_j^{(n)}}) = -\mathbb{E}(A_a^{(n)} \mid \mathcal{F}_{t_j^{(n)}})$.

$\implies Y_{t_j^{(n)}} = A_{t_j^{(n)}}^{(n)} - \mathbb{E}(A_a^{(n)} \mid \mathcal{F}_{t_j^{(n)}})$.

We will show $\{A_a^{(n)}\}_{n\geq 1}$ is U.I.. Then by Dunford-Pettis compactness criterion, $\exists$ integrable A_a

s.t.

$$A_a^{(n_k)} \xrightarrow{L^1} A_a$$

i.e. $\forall$ bounded r.v. ξ,

$$\lim_{k\to\infty} \mathbb{E}(\xi A_a^{(n_k)}) = \mathbb{E}(\xi A_a).$$

Define $A_t = Y_t + \mathbb{E}(A_a \mid \mathcal{F}_t)$.

4.11 Problem $\Rightarrow$ If $\mathbb{E}(\xi A_a^{(n_k)}) \to \mathbb{E}(\xi A_a)$ then

$$\mathbb{E}(\xi \mathbb{E}(A_a^{(n_k)} \mid \mathcal{F}_t)) \to \mathbb{E}(\xi \mathbb{E}(A_a \mid \mathcal{F}_t))$$

$A_a^{(n_k)} \xrightarrow{L^1} A_a \Rightarrow \ \mathbb{E}(A_a^{(n_k)} \mid \mathcal{F}_t) \xrightarrow{L^1} \mathbb{E}(A_a \mid \mathcal{F}_t)$.

$$\begin{cases} A_t = Y_t + \mathbb{E}(A_a \mid \mathcal{F}_t) \\ A_t^{(n)} = Y_t + \mathbb{E}(A_a^{(n)} \mid \mathcal{F}_t) \end{cases}$$

for any $t = t_j^{(n)}$.

Hence $\mathbb{E}(\xi A_t^{(n)}) \to \mathbb{E}(\xi A_t)$.

For any $0 \le s < t \le a$ with $s, t \in \{\frac{j}{2^n}, n \ge 1, 0 \le j \le 2^n\}$,

$$\mathbb{E}(\xi(A_t - A_s)) = \lim_{k\to\infty} \mathbb{E}(\xi(A_t^{(n_k)} - A_s^{(n_k)})) \ge 0 \quad \text{if } \xi \ge 0.$$

By letting $\xi = 1_{A_s > A_t}$, $\mathbb{E}\left[(A_t - A_s)1_{A_s - A_t}\right] \ge 0$.

$\Rightarrow \ A_t - A_s \ge 0$ a.s.

$A_0^{(n)} = 0 = Y_0 + \mathbb{E}(A_a^{(n)} \mid \mathcal{F}_0) \to Y_0 + \mathbb{E}(A_a \mid \mathcal{F}_0) = A_0$.

$\Rightarrow \ A_0 = 0$.

$X_t - \mathbb{E}(X_a \mid \mathcal{F}_t) = Y_t = A_t^{(n)} - \mathbb{E}(A_a^{(n)} \mid \mathcal{F}_t)$.

$\Rightarrow \ Y_t = A_t - \mathbb{E}(A_a \mid \mathcal{F}_t) = X_t - \mathbb{E}(X_a \mid \mathcal{F}_t)$.

$\Rightarrow \ X_t = A_t + \mathbb{E}(X_a - A_a \mid \mathcal{F}_t) = A_t + M_t$.

To prove $\{A_t\}$ is natural,

$$\mathbb{E}\int_0^a \xi_s \ dA_s = \mathbb{E}\int_0^a \xi_{s-} \ dA_s.$$

$$LHS = \mathbb{E}(\xi_a A_a) = \lim_{n\to\infty} \mathbb{E}(\xi_a A_a^{(n)})$$

$$RHS = \lim_{n\to\infty} \mathbb{E}\sum_{j=1}^{2^n} \xi_{t_{j-1}^{(n)}} (A_{t_j^{(n)}} - A_{t_{j-1}^{(n)}}).$$

Because $\{A^{(n)}_{t^{(n)}_j}\}_{0\le j\le 2^n}$ is predictable,

$$\begin{aligned}\mathbb{E}(\xi_a A^{(n)}_a) &= \mathbb{E}\left(\sum_{j=1}^{2^n} \xi_{t^{(n)}_{j-1}} (A^{(n)}_{t^{(n)}_j} - A^{(n)}_{t^{(n)}_{j-1}})\right)\\ &= \mathbb{E}\left(\sum_{j=1}^{2^n} \xi_{t^{(n)}_{j-1}} (Y_{t^{(n)}_j} - Y_{t^{(n)}_{j-1}})\right)\\ &= \mathbb{E}\left(\sum_{j=1}^{2^n} \xi_{t^{(n)}_{j-1}} (A_{t^{(n)}_j} - A_{t^{(n)}_{j-1}})\right)\end{aligned}$$

□

1.5 Square Integrable

Definition 1.5.1. Let $(X_t)_{t\ge 0}$ be a right-continuous martingale. We say X is square-integrable if

$$\mathbb{E}X_t^2 < \infty, \quad \forall\, t \ge 0.$$

Assume $X_0 = 0$, $X \in M_2$.

$Y_t = X_t^2$ is a submartingale.
Recall <u>DL</u>: $\forall$ stopping time $T \le a$, $\{Y_T\}_T$ is U.I..
By Doob-meyer decomposition, $Y_t = X_t^2 = M_t + A_t$, $\forall\, t \ge 0$.
If $X \in M_2^c$ (continuous), we may let M_t and A_t be continuous.

Definition 1.5.2. $\forall\, X \in M_2$, the quadratic variation of X is defined by $\langle X\rangle_t \triangleq A_t$ i.e. $\langle X\rangle_t$ is the unique adapted, natural increasing process such that $\langle X\rangle_0 = 0$ and $X_t^2 - \langle X\rangle_t$ is a martingale.

If we take $X, Y \in M_2$, then

$$\begin{cases}(X_t + Y_t)^2 - \langle X+Y\rangle_t \text{ is mart,}\\ (X_t - Y_t)^2 - \langle X-Y\rangle_t \text{ is mart.}\end{cases}$$

$\implies 4X_tY_t - (\langle X+Y\rangle_t - \langle X-Y\rangle_t)$ is a mart.

Definition 1.5.3. For $X, Y \in M_2$, we define the cross variation

$$\langle X, Y\rangle_t \triangleq \frac{1}{4}\left(\langle X+Y\rangle_t - \langle X-Y\rangle_t\right)$$

such that $X_tY_t - \langle X, Y\rangle_t$ is a martingale.

Remark. $\langle X, X\rangle_t = \langle X\rangle_t$.

Definition 1.5.4. The p-th variation of X over the partition $\Pi = \{t_0, \cdots, t_m\}$ of $[0, t]$ is

$$V_t^{(p)}(\Pi) = \sum_{k=1}^{m} \left|X_{t_k} - X_{t_{k-1}}\right|^p.$$

when $p = 2$, we call $V_t^{(2)}(\Pi)$ the quadratic variation.

Theorem 1.5.1. Let $X \in \mathcal{M}_2^c$. For partition Π of $[0, t]$, we have $V_t^{(2)}(\Pi) \xrightarrow{P} \langle X \rangle_t$.

Proof.

$$\begin{aligned}
&\mathbb{E}\left(V_t^{(2)}(\Pi) - \langle X \rangle_t\right)^2 \\
=&\mathbb{E}\left[\sum_{k=1}^{m} \left|X_{t_k} - X_{t_{k-1}}\right|^2 - \langle X \rangle_t\right]^2 \\
=&\mathbb{E}\left[\sum_{k=1}^{m} \left(\left|X_{t_k} - X_{t_{k-1}}\right|^2 - \left(\langle X \rangle_{t_k} - \langle X \rangle_{t_{k-1}}\right)\right)\right]^2 \\
=&\sum_{k=1}^{m} \mathbb{E}\left[\left(X_{t_k} - X_{t_{k-1}}\right)^2 - \left(\langle X \rangle_{t_k} - \langle X \rangle_{t_{k-1}}\right)\right]^2 \\
&+ 2 \sum_{1 \le j < \le m} \mathbb{E}\left[\left(X_{t_k} - X_{t_{k-1}}\right)^2 - \left(\langle X \rangle_{t_k} - \langle X \rangle_{t_{k-1}}\right)\right]\left[\left(X_{t_j} - X_{t_{j-1}}\right)^2 - \left(\langle X \rangle_{t_j} - \langle X \rangle_{t_{j-1}}\right)\right]. (\mathbf{H})
\end{aligned}$$

Notice that

$$\mathbb{E}\left[\left(X_{t_k} - X_{t_{k-1}}\right)^2 \middle| \mathcal{F}_{t_j}\right] = \mathbb{E}\left[X_{t_k}^2 \middle| \mathcal{F}_{t_j}\right] - 2\mathbb{E}\left(X_{t_k} X_{t_{k-1}} \middle| \mathcal{F}_{t_j}\right) + \mathbb{E}\left(X_{t_{k-1}}^2 \middle| \mathcal{F}_{t_j}\right).$$

For the cross-term,

$$\mathbb{E}\left(X_{t_k} X_{t_{k-1}} \middle| \mathcal{F}_{t_j}\right) = \mathbb{E}\left[\mathbb{E}\left(X_{t_k} X_{t_{k-1}} \middle| \mathcal{F}_{t_{k-1}}\right) \middle| \mathcal{F}_{t_j}\right] = \mathbb{E}\left[X_{t_{k-1}} \mathbb{E}\left(X_{t_k} \middle| \mathcal{F}_{t_{k-1}}\right) \middle| \mathcal{F}_{t_j}\right].$$

Hence

$$\mathbb{E}\left[\left(X_{t_k} - X_{t_{k-1}}\right)^2 \middle| \mathcal{F}_{t_j}\right] = \mathbb{E}\left[X_{t_k}^2 - X_{t_{k-1}}^2 \middle| \mathcal{F}_{t_j}\right] = \mathbb{E}\left[\langle X \rangle_{t_k} - \langle X \rangle_{t_{k-1}} \middle| \mathcal{F}_{t_j}\right]$$

Then

$$\begin{aligned}
(\mathbf{H}) &\le \sum_{k=1}^{m} \left[2\mathbb{E}\left(X_{t_k} - X_{t_{k-1}}\right)^4 + 2\mathbb{E}\left(\langle X \rangle_{t_k} - \langle X \rangle_{t_{k-1}}\right)^2\right] + 2\mathbb{E}\left[\sup_{1 \le k \le m} \left|\langle X \rangle_{t_k} - \langle X \rangle_{t_{k-1}}\right| \langle X \rangle_t\right] \\
&= I + II.
\end{aligned}$$

Note $I = 2\mathbb{E}V_t^{(4)}(\Pi)$.

Lemma 1.5.1. Let $X \in \mathcal{M}_2$ satisfy $|X_s| \le K < \infty$, $\forall\ s \le t$. Then $\mathbb{E}\left[V_t^{(2)}(\Pi)^2\right] \le 6K^4$

Assuming this, then

$$\mathbb{E}V_t^{(4)}(\Pi) \leq \mathbb{E}\left(\sum_{k=1}^{m} \left|X_{t_k} - X_{t_{k-1}}\right|^2 \cdot m_t^2\left(X; \|\Pi\|\right)\right)$$

where $m_t(X,\delta) = \sup\{|X_r - X_s| : \ |s-r| \leq \delta,\ 0 \leq r < s \leq t\}$. By Cauchy-Schwartz,

$$\begin{aligned}
&\mathbb{E}\left[V_t^{(4)}(\Pi)\right] \\
\leq &\mathbb{E}\left[V_t^{(2)}(\Pi) \cdot m_t^2(X, \|\Pi\|)\right] \\
\leq &\left(\mathbb{E}\left[V_t^{(2)}(\Pi)^2\right]\right)^{\frac{1}{2}} \left(\mathbb{E}m_t^4(X; \|\Pi\|)\right)^{\frac{1}{2}} \\
\leq &(6K^4)^{\frac{1}{2}} \cdot \left(\mathbb{E}\left[m_t^4\left(X, \|\Pi\|\right)\right]\right)^{\frac{1}{2}}.
\end{aligned}$$

If $|X_s| \leq K < \infty$, $\forall\ 0 \leq s \leq t$, by bounded convergence $\mathbb{E}\left[m_t^4(X, \|\Pi\|)\right] \to 0$ as $\|\Pi\| \to 0$.

Lemma 1.5.2. Let $X \in \mathcal{M}_2$ satisfy $|X_s| \leq K < \infty$, $\forall\ 0 \leq s \leq t$. Then

$$\mathbb{E}\left[V_t^{(4)}(\Pi)\right] \to 0 \quad \text{a.s.} \ \ \|\Pi\| \to 0.$$

For II. Let $X \in \mathcal{M}_2^c$ satisfy $\langle X\rangle_s \leq K < \infty$, $\forall\ 0 \leq s \leq t$. Then

$$\mathbb{E}\left[\langle X\rangle_t \cdot \sup_{1\leq k\leq m} \left|\langle X\rangle_{t_k} - \langle X\rangle_{t_{k-1}}\right|\right] \to 0.$$

[proof: By uniformly Continuity]

Summary: If $|X_s| \leq K < \infty$ and $\langle X\rangle_s \leq K$, $\forall\ 0 \leq s \leq t$, then

$$\mathbb{E}\left[\left(V_t^{(2)}(\Pi) - \langle X\rangle_t\right)^2\right] \to 0 \quad \text{a.s.} \ \ \|\Pi\| \to 0.$$

Next, for general $X \in \mathcal{M}_2^c$, we use "Localization" to define $\forall\ n \geq 1$,

$$T_n = \inf\{t \geq 0 \colon \ |X_t| \geq n \text{ or } \langle X\rangle_t \geq n\}.$$

Then $X_t^{(n)} = X_{t\wedge T_n}$ is a bounded martingale. Also $\left\{X_{t\wedge T_n}^2 - \langle X\rangle_{t\wedge T_n}\right\}_{t\geq 0}$ is a bounded mart.
By uniqueness of $\langle X^{(n)}\rangle_t$, we get $\langle X^{(n)}\rangle_t = \langle X\rangle_{t\wedge T_n}$ and $(X_t^{(n)})^2 - \langle X^{(n)}\rangle_t$ is mart.
Hence $\forall\ 0 \leq s \leq t$

$$\begin{cases} \left|X_s^{(n)}\right| \leq n < \infty, \\ \langle X^{(n)}\rangle_s \leq n < \infty. \end{cases}$$

$$\implies \mathbb{E}\left[\sum_{k=1}^{m} \left(X_{t_k\wedge T_n} - X_{t_{k-1}\wedge T_n}\right)^2 - \langle X\rangle_{t\wedge T_n}\right]^2 \to 0 \quad \text{a.s.} \ \ \|\Pi\| \to 0.$$

$T_n \uparrow \infty$ a.s. $\forall$ fixed $t > 0$, $\mathsf{P}(T_n < t) \to 0$. We need to prove $\forall\ \varepsilon > 0$, $\forall\ \eta > 0$, $\exists\ \delta > 0$, if

$\|\Pi\| \le \delta$, then

$$\mathsf{P}\left(\left|V_t^{(2)}(x) - \langle X\rangle_t\right| > \varepsilon\right) < \eta.$$

Let $n \ge 1$ be large such that $\mathsf{P}(T_n < t) \le \frac{\eta}{2}$. Then

$$\begin{aligned}\mathsf{P}\left(\left|V_t^{(2)}(X) - \langle X\rangle_t\right| > \varepsilon\right) &= \mathsf{P}(T_n < t) + \mathsf{P}\left(T_n \ge t, \left|V_{t\wedge T_n}^{(2)}(X) - \langle X\rangle_{t\wedge T_n}\right| > \varepsilon\right)\\ &\le \frac{\eta}{2} + \frac{1}{\varepsilon^2}\mathbb{E}\left[\left(V_{t\wedge T_n}^{(2)}(X) - \langle X\rangle_{t\wedge T_n}\right)^2\right] \quad (\mathbf{Q}).\end{aligned}$$

Let $\|\Pi\|$ be small such that

$$(\mathbf{Q}) \le \frac{\eta}{2} + \frac{1}{\varepsilon^2}\varepsilon^2\frac{\eta}{2}.$$

Hence $\mathsf{P}\left(\left|V_t^{(2)}(x) - \langle X\rangle_t\right| > \varepsilon\right) \le \eta$. □

Corollary 1.5.1. $V_t^{(2)}(\Pi) \xrightarrow{P} \langle X\rangle_t \implies \begin{cases}\forall\ q > 2, & V_t^{(q)}(\Pi) \xrightarrow{P} 0,\\ \forall\ q < 2, & V_t^{(q)}(\Pi) \xrightarrow{P} \infty\end{cases}$ on $\{\langle X\rangle_t > 0\}$.

$\mathcal{M}_2 = \{X\colon \mathbb{E}X_t^2 < \infty, \forall\ t\}$.
$\mathcal{M}^2 = \{X\colon \sup_{t\ge 0}\mathbb{E}X_t^2 < \infty\}$.
$\mathcal{M}_0^2 = \{X \in M^2\colon X_0 = 0\}$.

1.6 L^2Theory

The space $\mathcal{M}^2 \longleftrightarrow L^2(\mathcal{F}_\infty)$, $X \mapsto X_\infty$. $\forall\ X \in \mathcal{M}^2$, we have

$$\begin{cases}X_\infty = \lim_{t\to\infty} X_t \text{ a.s.},\\ X_t \xrightarrow{L^2} X_\infty,\end{cases}$$

$\forall\ X_\infty \in L^2(\mathcal{F}_\infty)$, we have $X_t =\triangleq \mathbb{E}\left(X_\infty|\mathcal{F}_t\right) \in \mathcal{M}^2$.

$$\mathbb{E}\left(X_t^2\right) = \mathbb{E}\left[\mathbb{E}\left(X_\infty|\mathcal{F}_t\right)^2\right] \le \mathbb{E}\left[\mathbb{E}\left(X_\infty^2|\mathcal{F}_t\right)\right] = \mathbb{E}[X_\infty^2].$$

$\forall\ X, Y \in \mathcal{M}^2$, we define $\langle X, Y\rangle \triangleq \mathbb{E}\left[X_\infty Y_\infty\right]$.

Theorem 1.6.1 (Doob's L^2 inequality)**.**

$$\mathbb{E}\left[\sup_{0\le s\le t}|X_s|^2\right] \le 4\mathbb{E}\left[X_t^2\right].$$

Let $t \to \infty$ to see

$$\mathbb{E}\left[\sup_{t\ge 0}|X_s|^2\right] \le 4\mathbb{E}[X_\infty^2].$$

$\mathcal{M}^2$ is complete: For any Cauchy sequence $\{M^{(n)}\}$ in $\mathcal{M}^2$, we have $\{M_\infty^{(n)}\}$ is Cauchy in $L^2(\mathcal{F}_\infty)$. $\implies \exists\ M_\infty \in L^2(\mathcal{F}_\infty)$ s.t. $M_\infty^{(n)} \to M_\infty$ in L^2.

Then if $M_t = \mathbb{E}(M_\infty \mid \mathcal{F}_t)$ then

$$\sup_{t\geq 0}\left|M_t^{(n)} - M_t\right| \to 0 \text{ in } L^2.$$

Proof.

$$\mathbb{E}\left[\sup_{t\geq 0}\left|M_t^{(n)} - M_t\right|^2\right] = \mathbb{E}\left[\sup_{t\geq 0}\mathbb{E}\left(M_\infty^{(n)} - M_\infty\middle|\mathcal{F}_t\right)^2\right] \leq 4\mathbb{E}\left[\left(M_\infty^{(n)} - M_\infty\right)^2\right] \to 0.$$

□

Corollary 1.6.1. The space $c\mathcal{M}^2$ of continuous mart in $\mathcal{M}^2$ is closed in the topology of $\mathcal{M}^2$.

Proof. If $M^{(n)} \to M$ for $M^{(n)} \in c\mathcal{M}^2$, then

$$\sup_{t\geq 0}\left|M_t^{(n)} - M_t\right| \to 0 \text{ in } L^2$$

$\exists\ (n_k)$ s.t. $\sup_{t\geq 0}\left|M_t^{(n_k)} - M_k\right| \to 0$ a.s.
$\implies (M_t)$ is continous. □

Definition 1.6.1 (Weakly orthogonal)**.** $\forall\ M, N \in \mathcal{M}_0^2$, if $\mathbb{E}[M_\infty N_\infty] = 0$.

Definition 1.6.2 (Strongly Orthogonal)**.** $\forall$ stopping time T, $\mathbb{E}[M_T N_T] = 0$.

Lemma 1.6.1. Strongly orthogonal if and only if MN is U.I.

Lemma 1.6.2. For any progressively measurable process such that $\forall$ stopping time T,

$$\begin{cases} \mathbb{E}\,|M_T| < \infty, \\ \mathbb{E}M_T = 0, \end{cases}$$

then M is U.I.

Definition 1.6.3 (stable subspaces of $\mathcal{M}_0^2$)**.** We say $\mathcal{N}_0 \subseteq \mathcal{M}_0^2$ is a stable subspace if

① N_0 is closed subspace,

② $\forall\ N \in \mathcal{N}_0$, $\forall$ stopping time T, we have $N^T \in \mathcal{N}_0$ ($N_t^T = N_{t\wedge T}$).

Example . $c\mathcal{M}_0^2$ is stable.

Theorem 1.6.2. Let $\mathcal{N}_0$ be a stable subspace of $\mathcal{M}_0^2$. Let $\mathcal{N}_0^\perp$ be the set of $Z \in \mathcal{M}_0^2$ which are weakly orthogonal to every element of $\mathcal{N}_0$, i.e.

$$\mathcal{N}_0^\perp = \{Z \in \mathcal{M}_0^2 \colon \langle Z, N\rangle = 0,\ \forall\ N \in \mathcal{N}_0\}.$$

Then $\mathcal{N}_0^\perp$ is a stable subspace of $\mathcal{M}_0^2$, and $\forall\ Z \in \mathcal{N}_0^\perp$, we have Z, N are strongly orthogonal for all $N \in \mathcal{N}_0$,

Finally, $\forall\ M \in \mathcal{M}_0^2$, $M = N + Z$.

Proof. $0 = \mathbb{E}(Z_n N) \xrightarrow{L^1} \mathbb{E}(ZN)$.

Next, $\forall\ Z \in \mathcal{N}_0^\perp$, then we have $\forall\ N \in \mathcal{N}_0$, $\mathbb{E}[Z_\infty N_\infty] = 0$.

For any stopping time T, we have $Z^T \in \mathcal{N}_0^\perp$ because

$$\begin{aligned}
\mathbb{E}[Z_\infty^T N_\infty] &= \mathbb{E}[Z_T N_\infty] \\
&= \mathbb{E}\left[Z_T \mathbb{E}\left(N_\infty | \mathcal{F}_T\right)\right] \quad = \mathbb{E}[Z_T N_T] \\
&= \mathbb{E}\left[\mathbb{E}\left(Z_\infty | \mathcal{F}_T\right) N_T\right] \\
&= \mathbb{E}[Z_\infty N_T] \\
&= \mathbb{E}[Z_\infty N_\infty^T].
\end{aligned}$$

By $N^T \in \mathcal{N}_0$, we get $\mathbb{E}[Z_\infty N_\infty^T] = 0$ by $Z \in \mathcal{N}_0^\perp$. We prove $Z^T \in \mathcal{N}_0^\perp \Rightarrow\ \mathcal{N}_0^\perp$ is stable. □

Definition 1.6.4. Define $d\mathcal{M}_0^2 \triangleq (c\mathcal{M}_0^2)^\perp$. $\forall\ M \in \mathcal{M}_0^2$, we write $M = M^c + M^d$ with $M^c \in c\mathcal{M}_0^2$, $M^d \in d\mathcal{M}_0^2$ and $M^c M^d$ is U.I. [by $\mathbb{E}\left[M_T^c M_T^d\right] = 0$.]

Define $(\Delta f)(t) = f(t) - f(t-)$.

Theorem 1.6.3 (Meyer)**.** Let $M \in \mathcal{M}_0^2$. Then $\exists$ unique increasing process $[M]$ null at 0 s.t.

① $M^2 - [M]$ is U.I.

② $\Delta[M] = (\Delta M)^2$.

Proof. $M_t^2 = (M_t^c + M_t^d)^2 = (M_t^c)^2 + 2M_t^c M_t^d + (M_t^d)^2$. □

For $M, N \in \mathcal{M}_0^2$, let $[M, N] \equiv \frac{1}{4}\left([M+N] - [M-N]\right)$.

Theorem 1.6.4 (Kunita & Watanabe)**.** The process $[M, N]$ is the unique FV. process null at 0 s.t.

① $MN - [M, N]$ is U.I.,

② $\Delta[M, N] = \Delta M \cdot \Delta N$.

1.7 Itó Integral

Let $M \in \mathcal{M}_0^2$.

If $H_t = Z \cdot 1_{(S,T]}(t)$ for some $Z \in \mathcal{F}_S$ and $S \le T$ (stopping times).

Define $\int_0^t H_s \, \mathrm{d}M_s \equiv (H \cdot M)(t) \equiv Z \cdot (M_{T\wedge t} - M_{S\wedge t})$.

We will prove $(H \cdot M) \in \mathcal{M}_0^2$.

① We will prove $\forall$ stopping time R satisfies

$$\begin{cases} \mathbb{E}\left|(H\cdot M)_R\right| < \infty \\ \mathbb{E}(H\cdot M)_R = 0. \end{cases}$$

Proof. Let $K > 0$ s.t. $|Z| \le K < \infty$. Then $\mathbb{E}\left|(H\cdot M)_R\right| \le K\mathbb{E}\left|M_{T\wedge R}\right| + K\mathbb{E}\left|M_{S\wedge R}\right|$.($\star 1$)

By optional stopping

$$\begin{aligned} \mathbb{E}\left|M_{T\wedge R}\right| &= \mathbb{E}\left[\left|\mathbb{E}\left(M_\infty|\mathcal{F}_{T\wedge R}\right)\right|\right] \\ &\le \mathbb{E}\left[\mathbb{E}\left(|M_\infty||\mathcal{F}_{T\wedge R}\right)\right] = \mathbb{E}\left|M_\infty\right|. \end{aligned}$$

So ($\star 1$) $\le 2K\mathbb{E}\left|M_\infty\right| < \infty$.

To prove $\mathbb{E}(H\cdot M)_R = 0 \Leftrightarrow \mathbb{E}\left[Z(M_{T\wedge R} - M_{S\wedge R})\right] = 0$.($\star 2$)

We may replace Z by 1_A for $A \in \mathcal{F}_S$. Then

$$\star 2 \Leftrightarrow \mathbb{E}\left[1_A(M_{T\wedge R} - M_{S\wedge R})\right] = 0.$$

Define

$$S_A \equiv \begin{cases} S, & \text{on } A, \\ \infty, & \text{on } A^C. \end{cases}$$

Then

$$\mathbb{E}\left[1_A(M_{T\wedge R} - M_{S\wedge R})\right] \Leftrightarrow \mathbb{E}\left[M_{T_A\wedge R} - M_{S_A\wedge R}\right] = 0.$$

□

Lemma 1.7.1. If $Z \in b\mathcal{F}_S$ and M is U.I. martingale. Then $H\cdot M$ is U.I. martingale.

Next, consider $H_t = \sum_{i=1}^n Z_i 1_{(S_i,T_i]}(t)$.

$(H\cdot M)_t = \sum_{i=1}^n Z_i(M_{T_i\wedge t} - M_{S_i\wedge t})$.

Corollary 1.7.1. If $Z_i \in b\mathcal{F}_{S_i}$ and M is U.I. martingale, then $(H\cdot M)_t$ is U.I. martingale.

Definition 1.7.1 (previsible process)**.** Previsible $\sigma-$algebra P on $(0,\infty)\times\Omega$ is defined to be the smallest σ-algebra s.t. every adapted L-process is P-measurable.

L-process: $(X_t)_{t>0}$ s.t. $X_t = X_{t-}$ & X_{t+} exists.

We say $(X_t)_{t>0}$ is previsible if it is P-measurable.

Lemma 1.7.2. Suppose S and T are stopping times with $S \le T \le \infty$. Let $Z \in b\mathcal{F}_S$ and $H = Z\cdot 1_{(S,T]}$. Then H is previsible.

Proof. $\forall\ t > 0$ and B Borel set, $\{H_t \in B\} \in \mathcal{F}_t$. We may assume $0 \notin B$.

Then $\{H_t \in B\} = \{Z \in B\} \cap \{S < t \le T\} = \{Z \in B\} \cap \{S < t\} \cap \{t \le T\}$.

Note that

$$\{S < t\} = \bigcup_{n=1}^\infty \{S \le t - \frac{1}{n}\}$$

.

Hence

$$\{H_t \in B\} = \bigcup_{n=1}^{\infty} \left(\{Z \in B\} \cap \{S \le t - \frac{1}{n}\} \right) \cap \{T \ge t\} \subseteq \mathcal{F}_{t-\frac{1}{n}} \subseteq \mathcal{F}_t.$$

□

Definition 1.7.2. $b\varepsilon = \left\{ \sum_{i=1}^{n} Z_i 1_{(S_i,T_i]} \colon n \ge 1, Z_i \in b\mathcal{F}_{S_i}, S_1 \le T_1 \le S_2 \le \cdots . \right\}$

claim: $\sigma(b\varepsilon) = \mathsf{P}$.

Proof. "$\subseteq$" done.

"$\supseteq$" For any bounded left-continuous process,

$$H_t = \lim_{k\uparrow\infty} \lim_{n\uparrow\infty} \sum_{i=2}^{n_k} H_{\frac{i-1}{n}} 1_{(\frac{i-1}{n},\frac{i}{n}]}(t).$$

Since H is adapted, we get $H_{\frac{i-1}{n}} \in b\mathcal{F}_{\frac{i-1}{n}}$.

Hence

$$\sum_{i=2}^{n_k} H_{\frac{i-1}{n}} 1_{(\frac{i-1}{n},\frac{i}{n}]}(t) \in b\varepsilon.$$

$\implies H_t \in \sigma(b\varepsilon)$. □

Lemma 1.7.3. Let $H \in b\varepsilon$, $M \in \mathcal{M}_0^2$. Then $H \cdot M \in \mathcal{M}_0^2$, and

$$\mathbb{E}(H \cdot M)_\infty^2 = \sum_{i=1}^{n} \mathbb{E}\left[Z_{i-1}^2 (M_{T_i} - M_{T_{i-1}})^2 \right]$$

if $H = \sum_{i=1}^{n} Z_{i-1} 1_{(T_{i-1},T_i]}$.

Proof. $(H \cdot M)_t =\equiv \sum_{i=1}^{n} Z_{i-1} \left(M_{T_i \wedge t} - M_{T_{i-1} \wedge t} \right)$. We have $(H \cdot M)$ is U.I. martingale.

Then $(H \cdot M)_t \to (H \cdot M)_\infty$ a.s. and in L^1, where $(H \cdot M)_\infty = \sum_{i=1}^{n} Z_{i-1} \Delta i$, $\Delta_i = M_{T_i} - M_{T_{i-1}}$.

Notice that

$$\mathbb{E}\left[(H \cdot M)_\infty^2 \right] = \mathbb{E} \sum_{i=1}^{n} Z_{i-1}^2 \Delta_i^2 + 2\mathbb{E} \left(\sum_{i<j} Z_{i-1} \Delta_i Z_{j-1} \Delta_j \right).$$

For $i \le j - 1 < j$, we get

$$\mathbb{E}\left[Z_{i-1} \Delta_i Z_{j-1} \Delta_j \right] = \mathbb{E}\left[Z_{i-1} \Delta_i \cdot Z_{j-1} \mathbb{E}\left[\Delta_j \middle| \mathcal{F}_{T_{j-1}} \right] \right] = 0.$$

By Doob's L^2 ineq.

$$\mathbb{E}\left[\sup_{t \ge 0} (H \cdot M)_t^2 \right] \le 4\mathbb{E}\left[(H \cdot M)_\infty^2 \right] < \infty.$$

$\implies H \cdot M \in \mathcal{M}_0^2$. □

Next, Recall that $M \in \mathcal{M}_0^2 \implies \exists$ quadratic variation $[M]$ s.t. $M^2 - [M]$ is U.I..

$$\mathbb{E}\left[(H \cdot M)_\infty^2\right] = \sum_{i=1}^{n} \mathbb{E}\left[Z_{i-1}^2 \left(M_{T_i} - M_{T_{i-1}}\right)^2\right] . \star 1$$

Note

$$\begin{aligned}\mathbb{E}\left[\left(M_{T_i} - M_{T_{i-1}}\right)^2 \middle| \mathcal{F}_{T_{i-1}}\right] &= \mathbb{E}\left[M_{T_i}^2 - M_{T_{i-1}}^2 \middle| \mathcal{F}_{T_{i-1}}\right] \\ &= \mathbb{E}\left[[M]_{T_i} - [M]_{T_{i-1}} \middle| \mathcal{F}_{T_{i-1}}\right].\end{aligned}$$

So

$$\star 1 \implies \sum_{i=1}^{n} \mathbb{E}\left[Z_{i-1}^2 \left([M]_{T_i} - [M]_{T_{i-1}}\right)\right] = \mathbb{E}\left[\int_0^\infty H_s^2 \,\mathrm{d}[M]_s\right].$$

$$\int_0^\infty \left(\sum_{i=1}^{\infty} Z_{i-1} \mathbf{1}_{(T_{i-1} - T_i]}\right)^2 \mathrm{d}[M]_s = \int_0^\infty \sum_{i=1}^{n} Z_{i-1}^2 \mathbf{1}_{(T_{i-1}, T_i]} \,\mathrm{d}[M]_s.$$

$\mathbb{E}(H \cdot M)_\infty^2 = \mathbb{E}\left[\int_0^\infty H_s^2 \,\mathrm{d}[M]_s\right] < \infty$.
For any fixed $M \in \mathcal{M}_0^2$, define

$$\|H\|_M = \left(\mathbb{E}\int_0^\infty H_s^2 \,\mathrm{d}[M]_s\right)^{\frac{1}{2}}.$$

Then $L^2(M) \equiv \{\text{previsible } H \text{ with } \|H\|_M < \infty\}$.
We have

$$\begin{aligned} I : L^2(M) \cap b\varepsilon &\to L^2(\mathcal{F}_\infty) \\ H &\mapsto (H \cdot M)_\infty. \end{aligned}$$

$\|H\|_M = \left(\mathbb{E}\left[\int_0^\infty H_s^2 \,\mathrm{d}[M]_s\right]\right)^{\frac{1}{2}}$.
$\|I(H)\| = \left(\mathbb{E}(H \cdot M)_\infty^2\right)^{\frac{1}{2}}$.
And $\|H\|_M = \|I(H)\|$, we get isometry:

$$\mathbb{E}\left[\int_0^\infty H_s^2 \,\mathrm{d}[M]_s\right] = \mathbb{E}(H \cdot M)_\infty^2.$$

$\implies$ exstends uniquely to $L^2(M) \cap b\varepsilon = L^2(M)$.

Definition 1.7.3 (Itó Integral)**.** Let $M \in \mathcal{M}_0^2$ and $H \in L^2(M)$. Then Itó Integral $(H \cdot M)_t$ is the image of H under extension of $I : L^2(M) \to L^2(\mathcal{F}_\infty)$.

Localization

Definition 1.7.4. Let T be a stopping time and H is any process. Define

$$H(0, T](t) = \begin{cases} H_t, & 0 < t \le T, \\ 0, & t > T. \end{cases}$$

Remark. If H is previsible, so is $H(0,T]$.

Definition 1.7.5. Let T be a stopping time and X is any process. Define

$$X^T(t) = \begin{cases} X(t), & 0 \le t \le T, \\ X(T), & t > T. \end{cases}$$

Then $(H \cdot M)^T = (H(0,T] \cdot M^T)_t$, and

$$\left[Z_i \left(M_{T_i \wedge t} - M_{T_{i-1} \wedge t}\right)\right]^T = \left(Z_i \left(T_{i-1} \wedge T, T_i \wedge T\right] \cdot M^T\right)_t = Z_i \left(M_{T_i \wedge T \wedge t} - M_{T_{i-1} \wedge T \wedge t}\right).$$

Theorem 1.7.1. Fix $M \in \mathcal{M}_0^2$. For any $H \in L^2(M)$ and any stopping time T.

(i) $(H \cdot M)^T = H(0,T] \cdot M = H \cdot M^T$.

(ii) $(H \cdot M)_t^2 - \int_0^\infty H_s^2 \, \mathrm{d}[M]_s$ is U.I. martingale. $\implies [H \cdot M]_t = \int_0^t H_s^2 \, \mathrm{d}[M]_s$.

(iii) If $H, K \in b\mathsf{P}$, then $(HK) \cdot M = H \cdot (K \cdot M) = K \cdot (H \cdot M)$.

(iv) $\Delta(H \cdot M) = H \cdot \Delta M$.

(v) If M is cts, then $H \cdot M$ is cts.

Proof. (i) $(H \cdot M)_t^T = (H \cdot M)_{t \wedge T}$. □

Example . $H = Z1_{(U,V]}$.
$(H \cdot M)_t = Z(M_{V \wedge t} - M_{U \wedge t}) \implies (H \cdot M)_{t \wedge T} = Z\left(M_{V \wedge T \wedge t} - M_{U \wedge T \wedge t}\right)$.
$H(0,T](t) = Z1_{(U \wedge T, V \wedge T]}(t)$.
So $(H(0,T] \cdot M)_t = Z(M_{V \wedge T \wedge t} - M_{U \wedge T \wedge t})$.
$(H \cdot M^T)_t = Z(M_{V \wedge t}^T - M_{U \wedge t}^T) = Z(M_{T \wedge V \wedge T} - M_{T \wedge U \wedge t})$.

Hence it holds for $H \in b\varepsilon$. $\implies$ holds for all $H \in L^2(M)$.
$(H \cdot M)_\infty^T = Z(M_V - M_U)$.

Proof. (ii) $N_t = (H \cdot M)_t^2 - \int_0^t H_s^2 \, \mathrm{d}[M]_s$.

$$\mathbb{E}\left[\sup_{t \ge 0} (H \cdot M)_t^2\right] \le 4\mathbb{E}\left[\int_0^\infty H_s^2 \, \mathrm{d}[M]_s\right] < \infty.$$

Hence $\mathbb{E}\left[\sup_{t \ge 0} |N_t|\right] < \infty \implies \{N_t, t \ge 0\}$ is U.I..
It suffices to prove $\forall$ stopping time T,

$$\begin{cases} \mathbb{E}[N_T] = 0 \\ \mathbb{E}\,|N_T| < \infty \end{cases} \implies \text{U.I. martingale.}$$

The second condition is proved, Now we need to check the first condition.

$$\mathbb{E}[N_T] = \mathbb{E}\left[(H\cdot M)_T^2\right] - \mathbb{E}\int_0^T H_s^2 \,\mathrm{d}[M]_s.$$

Notice that $(H\cdot M)_T = Z(M_{V\wedge T} - Z_{U\wedge T}) = (H\cdot M)_\infty^T$.

Then

$$(H\cdot M)_T = (H\cdot M)_\infty^T = (H(0,T]\cdot M)_\infty$$

$$\implies \mathbb{E}\left[(H\cdot M)_T^2\right] = \mathbb{E}\left[(H(0,T]\cdot M)_\infty^2\right] = \mathbb{E}\left[\int_0^\infty (H(0,T]_s)^2 \,\mathrm{d}[M]_s\right] = \mathbb{E}\left[\int_0^T H_s^2 \,\mathrm{d}[M]_s.\right]$$

□

Proof. (iii) $H = Z_1(S_1,T_1]$, $K = Z_2(S_2,T_2]$.

$HK = Z_1Z_2(S_1\vee S_2,(T_1\wedge T_2)\vee(S_1\vee S_2)]$.

$$((HK)\cdot M)_\infty = Z_1Z_2\left(M_T - M_S\right).$$

$$\begin{aligned}
(H\cdot(K\cdot M))_\infty &= Z_1\left((K\cdot M)_{T_1} - (K\cdot M)_{S_1}\right)\\
&= Z_1\left[(K\cdot M)_\infty^{T_1} - (K\cdot M)_\infty^{S_1}\right]\\
&= Z_1\left[(K\cdot M^{T_1})_\infty - (K\cdot M^{S_1})_\infty\right]\\
&= Z_1\left[Z_2\left(M_{T_2}^{T_1} - M_{S_2}^{T_1}\right) - Z_2\left(M_{T_2}^{S_1} - M_{S_2}^{S_1}\right)\right]\\
&= Z_1Z_2\left[M_{T_2\wedge T_1} - M_{S_2\wedge T_1} - (M_{T_2\wedge S_1} - M_{S_2\wedge S_1})\right]\\
&= Z_1Z_2\left(M_T - M_S\right).
\end{aligned}$$

So $H\cdot(K\cdot M) = (H\cdot K)\cdot M$, $\forall\ H,K\in b\varepsilon$.

Next, for any $k\in b\mathsf{P}$ s.t. $\exists\ k^n\in b\varepsilon$, $k^n \xrightarrow{L^2(M)} k$.

Then $\forall\ H\in b\varepsilon$,

$$\mathbb{E}\left((H\cdot(K\cdot M) - H\cdot(k^n\cdot M))_\infty^2\right) = \mathbb{E}\left((H\cdot(k-k^n)\cdot M)_\infty^2\right) = \mathbb{E}\left[\int_0^\infty H_s^2 \,\mathrm{d}[(k-k^n)\cdot M]_s\right].$$

Recall:

$$[H\cdot M]_t = \int_0^t H_s^2 \,\mathrm{d}[M]_s = \mathbb{E}\left[\int_0^\infty H_s^2(k_s-k_s^n)^2 \,\mathrm{d}[M]_s\right] \to 0,$$

by $|H_s|\le C$ and $k^n \xrightarrow{L^2(M)} k$.

$$\|k^n - k\|_M^2 \triangleq \mathbb{E}\left[\int_0^\infty (k_s-k_s^n)^2 \,\mathrm{d}[M]_s\right] \to 0.$$

Hence $\mathbb{E}\left[((H\cdot(k^n\cdot M)) - H\cdot(K\cdot M))_\infty^2\right] \to 0$.

Next

$$\mathbb{E}\left[H\cdot(k^n\cdot M)^2_\infty\right]=\mathbb{E}\left[((H\cdot k^n)\cdot M)^2_\infty\right]\to\mathbb{E}\left[((HK)\cdot M)^2_\infty\right].$$

We conclude

$$\mathbb{E}\left[(H\cdot(k\cdot M))^2_\infty\right]=\mathbb{E}\left[((Hk)\cdot M)^2_\infty\right].$$

□

Proof. (iv) If $H=Z(S,T]$, then $(H\cdot M)_t=Z\left(M_{t\wedge T}-M_{t\wedge S}\right)$.

$$\Delta(H\cdot M)_t=\Delta(Z(M_t-M_S))=Z\times\Delta M_t=H\times\Delta M_t.$$

$\Longrightarrow$ holds for all $H\in b\varepsilon$.
Next, $\forall\ H\in L^2(M)$, we let $H^n\in b\varepsilon$ and $H^n\xrightarrow{L^2(M)}H$. Then

$$H^n\cdot M\xrightarrow{L^2(\mathcal{F}_\infty)}H\cdot M\Longleftrightarrow H^n\cdot M\xrightarrow{\mathcal{M}_0^2}H\cdot M.$$

$\mathbb{E}\left[\int_0^\infty(H^n_S-H_S)^2\,\mathrm{d}[M]_s\right]\to 0\implies\exists$ subseq $n_k\to\infty$ s.t. W.P.1.

$$\int_0^\infty(H^n_S-H_S)^2\,\mathrm{d}[M]_s\to 0.$$

$$\sup_{t\geq 0}|(H^n\cdot M-H\cdot M)_t|\xrightarrow{L^2}0\implies\sup_{t\geq 0}|(H^{n_k}\cdot M)_t-(H\cdot M)_t|\xrightarrow{\text{a.s.}}0.$$

By $\Delta(H^{n_k}\cdot M)_t=H^{n_k}\cdot\Delta M_t$.
case 1. If $\Delta M_S=0$, then $\Delta(H^{n_k}\cdot M)_S=0$.
By uniform convergence $\Delta(H\cdot M)_S=0$.
case 2. If $\Delta M_S\neq 0$, then $\Delta[M]_S=(\Delta M_S)^2>0$.
Then $H^{n_k}_S-H_s\to 0$ by

$$0\leftarrow\int_0^\infty\left(H^{n_k}_t-H_t\right)^2\mathrm{d}[M]_t\geq\left(H^{n_k}_S-H_S\right)^2\cdot\Delta[M]_S.$$

$\Delta(H^{n_k}\cdot M)_S=H^{n_k}_S\cdot\Delta M_S$, let $k\to\infty$ we have

$$\Delta(H\cdot M)_S=H_S\cdot\Delta M_S.$$

□

Definition 1.7.6 (defn example)**.** Example

Theorem 1.7.2 (thm example)**.** Example

Proof. proof □

Lemma 1.7.4 (lemma example)**.** Example

Proof for Lemma.

■. proof for lemma example □

Proposition 1.7.1. pro example

Proof for Proposition.

■. Example □

Claim 1.7.1. claim example Example

Proof for Claim.

■. proof for claim example □

Corollary 1.7.2 (cor example)**.** example

Proof for Corollary.

■. proof for corollary example □

Fact 1.7.1. Example fact

Remark. Example remark

zhengzh2025@mail.sustech.edu.cn